Selim El Fata

AI Engineering Student · Software Developer

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Fourth-year AI Engineering student who designs, builds and ships real software. I delivered a clinic management app that a dentist in Ghana now uses every day, and built a real-time black hole simulator that ray traces light through curved spacetime on the GPU. Open to freelance projects and full-time or part-time roles.

EXPERIENCE

Freelance Software Developer · Pro-denticare, dental clinic, Ghana — 2026

- Designed and built a desktop clinic management app, delivered to the client and in daily use: patient records, appointment calendar, treatment records, invoices, receipts, prescriptions and patient email.
- Built an interactive 3D dental chart in Three.js with an X-ray view, panoramic mode and a playable treatment-history timeline.
- Made it work fully offline with a local SQLite database, scheduled backups and migrations tested against every past version; packaged installers for macOS and Windows and shipped v1.2.
- Designed and built the product's marketing website.

Electron · React · TypeScript · SQLite · Three.js · Tailwind · Zustand · Zod

Software Development Intern · Melanion SAL — Jun 2026

- Completed a 40-hour software development program covering C# and object-oriented design, Git collaborative workflows, REST APIs and JSON, SQL data modelling and queries, and data cleaning, analysis and visualisation.

Cashier · Treasure Island — 2023 – 2025

- Served a high volume of customers and handled payments accurately in a fast-paced environment, building strong communication skills and the ability to stay calm and reliable under pressure.

PROJECTS

Event Horizon · real-time black hole simulator — selim7417.github.io/event-horizon

- Wrote a GPU ray tracer for Schwarzschild spacetime from scratch, following every pixel's light ray backwards in real time; the black hole's shadow matches theory to within 0.35%.
- Physically based colour from Planck's law and CIE 1931, with Doppler beaming and gravitational redshift; dynamic resolution, temporal anti-aliasing and an eight-chapter guided tour.

WebGL2 · GLSL · JavaScript · no libraries

Personal Portfolio · animated portfolio website — selim7417.github.io

- Designed and built an animated, responsive portfolio with an interactive particle canvas and scroll-driven motion, tuned to a steady 60 fps with no frameworks.

EDUCATION

Artificial Intelligence Engineering · ESIB, Université Saint-Joseph (USJ) — Expected 2028

Currently in 4th year of a 5-year engineering programme · Beirut, Lebanon

SKILLS

Languages	Python · C++ · C# · JavaScript · TypeScript · SQL · GLSL · HTML/CSS
Frameworks	React · Electron · Three.js · WebGL2 · Tailwind CSS · Zustand · Zod
Tools	Git & GitHub · SQLite · REST APIs · Vite · Object-oriented design & design patterns
Spoken	Arabic (excellent) · English (excellent) · French (very good)

STRENGTHS

End-to-end ownership
Took a client's app from first sketch to daily use, including design, code, testing and delivery.

Calm under pressure
Two years of customer-facing work during busy shifts, staying accurate and friendly.

Fast learner
Taught myself GPU programming and 3D graphics to build a physics-accurate simulator from scratch.

Clear communicator
Explains technical work in plain words and works directly with clients to understand what they need.